

ANSWERS

1.

(a) e.g. separate charges, store energy, smoothing circuit. etc. B1
(allow 'stores charge')

(b) (i) charge = current $\times$ time B1

(ii) area is 21.2 cm^2 (allow $\pm 0.5 \text{ cm}^2$) C2
(allow 1 mark if outside $\pm 0.5 \text{ cm}^2$ but within $\pm 1.0 \text{ cm}^2$)

1.0 cm^2 represents $(0.125 \times 10^{-3} \times 1.25 =) 156 \mu\text{C}$ C1

charge = $3300 \mu\text{C}$ A1

(iii) capacitance = Q/V C1

$$= (3300 \times 10^{-6}) / 15$$

$$= 220 \mu\text{F} \text{ A1}$$

(c) either energy = $\frac{1}{2}CV^2$ or energy = $\frac{1}{2}QV$ and $C = Q/V$ C1

$$\frac{1}{2} \times C \times 15^2 = 2 \times \frac{1}{2} \times C \times V^2 \text{ C1}$$

$$V = 10.6 \text{ V} \text{ A1}$$

2.

(a) at $t = 1.0 \text{ s}$, $V = 2.5 \text{ V}$ C1

$$\text{energy} = \frac{1}{2}CV^2 \text{ C1}$$

$$0.13 = \frac{1}{2} \times C \times (8.0^2 - 2.5^2) \text{ M1}$$

$$C = 4500 \mu\text{F} \text{ A0}$$

(b) use of two capacitors in series in all branches of combination M1
connected into correct parallel arrangement A1

3.

(a) (i) ratio of charge (on body) and its potential B1
(do not allow reference to plates of a capacitor)

(ii) (potential at surface of sphere =) $V = Q / 4\pi\epsilon_0 r$ M1

$$C = Q / V = 4\pi\epsilon_0 r \text{ A0}$$

(b) (i) $C = 4 \times \pi \times 8.85 \times 10^{-12} \times 0.36$
 $= 4.0 \times 10^{-11} \text{ F}$ (allow 1 s.f.) A1

(ii) $Q = CV$
 $= 4.0 \times 10^{-11} \times 7.0 \times 10^5$
 $= 2.8 \times 10^{-5} \text{ C} \text{ A1}$

(c) plastic is an insulator / not a conductor / has no free electrons B1
 charges do not move (on an insulator) B1
either so no single value for the potential
or charge cannot be considered to be at centre B1

(d) *either* energy = $\frac{1}{2}CV^2$ *or* energy = $\frac{1}{2}QV$ and $C = Q/V$ C1
 energy = $\frac{1}{2} \times 4 \times 10^{-11} \times \{(7.0 \times 10^5)^2 - (2.5 \times 10^5)^2\}$ C1
 = 8.6 J A1

4.

(a) charge / potential(ratio must be clear) B1

(b) potential (at surface of sphere) = $Q / 4\pi\epsilon_0 R$ M1
 $C = Q / V = 4\pi\epsilon_0 R$ A0

(c) (i) $C = 4\pi \times 8.85 \times 10^{-12} \times 0.63$ C1
 = 7.0×10^{-11} A1
 farad / FB1

(ii) energy = $\frac{1}{2}CV^2$ C1
 $0.25 \times \frac{1}{2}C \times (1.2 \times 10^6)^2 = \frac{1}{2}CV^2$ C1
 $V = 6.0 \times 10^5$ VA1
 (use of 0.75 rather than 0.25, allow max 2 marks)

5.

(a) e.g. 'storage of charge' / storage of energy
 blocking of direct current
 producing of electrical oscillations
 smoothing
 (any two, 1 mark each) B2

(b) (i) capacitance of parallel combination = 60 μ F C1
 total capacitance = 20 μ F A1

(ii) p.d. across parallel combination = $\frac{1}{2} \times$ p.d. across single capacitor C1
 maximum is 9V A1

(c) *either* energy = $\frac{1}{2}CV^2$ *or* energy = $\frac{1}{2}QV$ and $Q = CV$ C1
 energy = $\frac{1}{2} \times 4700 \times 10^{-6} \times (18^2 - 12^2)$ C1
 = 0.42 J A1

6.

(a) charge / potential (difference) (ratio must be clear) B1

(b) (i) $V = Q / 4\pi\epsilon_0 r$ B1

(ii) $C = Q / V = 4\pi\epsilon_0 r$ and $4\pi\epsilon_0$ is constant M1
so $C \propto r$ A0

(c) (i) $r = C / 4\pi\epsilon_0$ C1
 $r = (6.8 \times 10^{-12}) / (4\pi \times 8.85 \times 10^{-12})$ C1
 $= 6.1 \times 10^{-2} \text{ m}$ A1

(ii) $Q = CV = 6.8 \times 10^{-12} \times 220$
 $= 1.5 \times 10^{-9} \text{ C}$ A1

(d) (i) $V = Q/C = (1.5 \times 10^{-9}) / (18 \times 10^{-12})$
 $= 83 \text{ V}$ A1

(ii) either energy = $\frac{1}{2}CV^2$ C1
 $\Delta E = \frac{1}{2} \times 6.8 \times 10^{-12} \times 220^2 - \frac{1}{2} \times 18 \times 10^{-12} \times 83^2$ C1
 $= 1.65 \times 10^{-7} - 6.2 \times 10^{-8}$
 $= 1.03 \times 10^{-7} \text{ J}$ A1
or energy = $\frac{1}{2}QV$ (C1)
 $\Delta E = \frac{1}{2} \times 1.5 \times 10^{-9} \times 220 - \frac{1}{2} \times 1.5 \times 10^{-9} \times 83$ (C1)
 $= 1.03 \times 10^{-7} \text{ J}$ (A1)

7.

(a) (i) work done moving unit positive charge M1
from infinity to the point A1

(ii) charge / potential (difference) (ratio must be clear) B1

(b) (i) capacitance = $(2.7 \times 10^{-6}) / (150 \times 10^3)$ C1
(allow any appropriate values)
capacitance = 1.8×10^{-11} (allow 1.8 ± 0.05) A1

(ii) either energy = $\frac{1}{2}CV^2$ or energy = $\frac{1}{2}QV$ and $Q = CV$ C1
energy = $\frac{1}{2} \times 1.8 \times 10^{-11} \times (150 \times 10^3)^2$ or $\frac{1}{2} \times 2.7 \times 10^{-6} \times 150 \times 10^3$
 $= 0.20 \text{ J}$ A1

(c) either since energy $\propto V^2$, capacitor has $(\frac{1}{2})^2$ of its energy left
or full formula treatment C1
energy lost = 0.15 J A1

8.

- (a) charges on plates are equal and opposite M1
 so no resultant charge A1
 energy stored because there is charge separation B1
- (b) (i) capacitance = Q / V C1
 $= (18 \times 10^{-3}) / 10$
 $= 1800 \mu\text{F}$ A1
- (ii) use of area under graph or energy = $\frac{1}{2}CV^2$ C1
 energy = $2.5 \times 15.7 \times 10^{-3}$ or energy = $\frac{1}{2} \times 1800 \times 10^{-6} \times (10^2 - 7.5^2)$
 $= 39\text{mJ}$ A1
- (c) combined capacitance of Y & Z = $20 \mu\text{F}$ or total capacitance = $6.67 \mu\text{F}$ C1
 p.d. across capacitor X = 8V or p.d. across combination = 12V C1
 charge = $10 \times 10^{-6} \times 8$ or $6.67 \times 10^{-6} \times 12$
 $= 80 \mu\text{C}$ A1

9.

- (a) e.g. storing energy
 separating charge
 blocking d.c.
 producing electrical oscillations
 tuning circuits
 smoothing
 preventing sparks
 timing circuits
(any two sensible suggestions, 1 each, max 2) B2
- (b) (i) $-Q$ (induced) on opposite plate of C_1 B1
 by charge conservation, charges are $-Q, +Q, -Q, +Q, -Q$ B1
- (ii) total p.d. $V = V_1 + V_2 + V_3$ B1
 $Q/C = Q/C_1 + Q/C_2 + Q/C_3$ B1
 $1/C = 1/C_1 + 1/C_2 + 1/C_3$ A0
- (c) (i) energy = $\frac{1}{2}CV^2$ or energy = $\frac{1}{2} QV$ and $C = Q/V$ C1
 $= \frac{1}{2} \times 12 \times 10^{-6} \times 9.0^2$
 $= 4.9 \times 10^{-4} \text{ J}$ A1
- (ii) energy dissipated in (resistance of) wire/as a spark B1

10.

- (a) (i) ratio of charge and potential (difference)/voltage
(ratio must be clear) B1
- (ii) capacitor has equal magnitudes of (+)ve and (-)ve charge B1
total charge on capacitor is zero (so does not store charge) B1
 (+)ve and (-)ve charges to be separated M1
 work done to achieve this so stores energy A1
- (b) (i) capacitance of Y and Z together is $24 \mu\text{F}$ C1
 $1/C = 1/24 + 1/12$
 $C = 8.0 \mu\text{F}$ (allow 1 s.f.) A1
- (ii) some discussion as to why all charge of one sign on one plate of X B1
 $Q = (CV =) \underline{8.0 \times 10^{-6}} \times 9.0$ M1
 $= 72 \mu\text{C}$ A0
- (iii) 1. $V = (72 \times 10^{-6}) / (12 \times 10^{-6})$
 $= 6.0 \text{ V}$ (allow 1 s.f.) (allow 72/12) A1
2. either $Q = 12 \times 10^{-6} \times 3.0$ or charge is shared between Y and Z C1
 charge = $36 \mu\text{C}$ A1
 Must have correct voltage in (iii)1 if just quote of $36 \mu\text{C}$ in (iii)2.

11.

- (a) (i) $V = q / 4\pi\epsilon_0 R$ B1
- (ii) (capacitance is) ratio of charge and potential or q/V M1
 $C = q/V = 4\pi\epsilon_0 R$ A0
- (b) (i) $C = 4\pi \times 8.85 \times 10^{-12} \times 0.45$ C1
 $= 50 \text{ pF}$ A1
- (ii) either energy = $\frac{1}{2} CV^2$ or energy = $\frac{1}{2} QV$ and $Q = CV$ C1
 energy of spark = $\frac{1}{2} \times 50 \times 10^{-12} \{(9.0 \times 10^5)^2 - (3.6 \times 10^5)^2\}$ C1
 $= 17 \text{ J}$ A1

12.

- (a) e.g. store energy (do not allow 'store charge')
 in smoothing circuits
 blocking d.c.
 in oscillators
any sensible suggestions, one each, max. 2

B2

- (b) (i) potential across each capacitor is the same and $Q = CV$

B1

- (ii) total charge $Q = Q_1 + Q_2 + Q_3$

M1

$$CV = C_1V + C_2V + C_3V$$

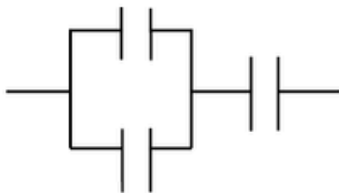
M1

(allow $Q = CV$ here or in (i))

$$\text{so } C = C_1 + C_2 + C_3$$

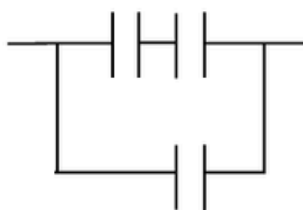
A0

- (c) (i)



A1

- (ii)



A1

13.

- (a) (i) energy = EQ
 $= 9.0 \times 22 \times 10^{-3}$
 $= 0.20 \text{ J}$

C1

A1

- (ii) 1. $C = Q/V$

$$V = (22 \times 10^{-3}) / (4700 \times 10^{-6})$$

$$= 4.7 \text{ V}$$

C1

A1

2. either $E = \frac{1}{2}CV^2$
 $= \frac{1}{2} \times 4700 \times 10^{-6} \times 4.7^2$
 $= 5.1 \times 10^{-2} \text{ J}$

C1

A1

or $E = \frac{1}{2}QV$
 $= \frac{1}{2} \times 22 \times 10^{-3} \times 4.7$
 $= 5.1 \times 10^{-2} \text{ J}$

(C1)

(A1)

or $E = \frac{1}{2}Q^2/C$
 $= \frac{1}{2} \times (22 \times 10^{-3})^2 / 4700 \times 10^{-6}$
 $= 5.1 \times 10^{-2} \text{ J}$

(C1)

(A1)

(b) energy lost (as thermal energy) in resistance/wires/battery/resistor B1
(award only if answer in (a)(i) > answer in (a)(ii)2)

14.

(a) for the two capacitors in parallel, capacitance = 96 μF C1
for complete arrangement, $1/C_T = 1/96 + 1/48$
 $C_T = 32 \mu\text{F}$ A1

(b) p.d. across parallel combination is one half p.d. across single capacitor C1
total p.d. = 9 V A1

15.

(a) (capacitance =) charge/potential (difference) B1

(b) $V = V_1 + V_2 + V_3$ B1
either $Q/C = Q/C_1 + Q/C_2 + Q/C_3$ or $V/Q = V_1/Q + V_2/Q + V_3/Q$
and so $1/C = 1/C_1 + 1/C_2 + 1/C_3$ B1

(c) (i) 1. $1/C_T = (1/200) + (1/600)$
 $C_T = 150 \mu\text{F}$ A1

2. $Q = CV$
 $= 150 \times 10^{-6} \times 12$ or $600 \times 10^{-6} \times 3.0$ or $200 \times 10^{-6} \times 9.0$
 $= 1.8 \times 10^{-3} \text{ C}$ A1

3. $V = Q/C = 1.8 \times 10^{-3} / 600 \times 10^{-6}$ or $V = [200 / (200 + 600)] \times 12$
 $= 3(.0) \text{ V}$ A1

(ii) energy = $\frac{1}{2}CV^2$ or energy = $\frac{1}{2}QV$ and $C = Q/V$ C1
 $\frac{1}{2} \times C \times 3^2 = 2 \times \frac{1}{2} \times C \times V^2$ C1
 $V = 2.1 \text{ V}$ A1

16.

(a) e.g. storing energy
blocking d.c.
in oscillator circuits
in tuning circuits
in timing circuits

any two B2

(b) (i)	$1/6 + 1/C + 1/C = 1/4$	C1
	$C = 24\ \mu\text{F}$	A1
(ii)	$Q = CV$	
	$= 4.0 \times 10^{-6} \times 12$	C1
	$= 48\ \mu\text{C}$	A1
(iii)	1. $48\ \mu\text{C}$	A1
	2. $24\ \mu\text{C}$	A1
17.		
(a) (i)	charge / potential (difference) or charge per (unit) potential (difference)	B1
(ii)	$(V = Q/4\pi\epsilon_0 r \text{ and } C = Q/V)$	
	for sphere, $C (= Q/V) = 4\pi\epsilon_0 r$	C1
	$C = 4\pi \times 8.85 \times 10^{-12} \times 12.5 \times 10^{-2} = 1.4 \times 10^{-11}\text{ F}$	A1
(b) (i)	$1/C_T = 1/3.0 + 1/6.0$	
	$C_T = 2.0\ \mu\text{F}$	A1
(ii)	total charge = charge on $3.0\ \mu\text{F}$ capacitor = $2.0\ (\mu) \times 9.0 = 18\ (\mu\text{C})$	C1
	potential difference = $Q/C = 18\ (\mu)\text{C} / 3.0\ (\mu)\text{F} = 6.0\text{ V}$	A1
	or	
	argument based on equal charges:	
	$3.0 \times V = 6.0 \times (9.0 - V)$	(C1)
	$V = 6.0\text{ V}$	(A1)
(iii)	potential difference (= $9.0 - 6.0$) = 3.0 V	C1
	charge (= $3.0 \times 2.0\ (\mu)$) = $6.0\ \mu\text{C}$	A1

18.

7(a)	equal and opposite charges on the plates so no resultant charge	B1
	+ve and -ve charges separated so energy stored	B1
7(b)	charge / potential difference	M1
	reference to charge on one plate and p.d. between plates	A1
7(c)	energy = $\frac{1}{2} CV^2$ or energy = $\frac{1}{2} QV$ and $C = Q / V$	C1
	$(1 / 16) \times \frac{1}{2} CV_0^2 = \frac{1}{2} CV^2$ $V = \frac{1}{4} V_0$	A1

19.

6(a)(i)	$1/T = 1/(2C) + 1/C$	C1
	$T = \frac{2}{3}C$ or $0.67C$	A1
6(a)(ii)	same charge on Q as on combination	B1
	so p.d. is 6.0 V	B1
6(b)	P: p.d. will decrease (from 3.0V)	B1
	to zero	B1
	Q: p.d. will increase (from 6.0V)	B1
	to 9.0V	B1

20.

7(a)	(capacitance =) charge / potential	M1
	charge is (numerically equal to) charge on one plate	A1
	potential is potential difference between plates	A1
7(b)(i)	$4.5 \times 10^{-6} \text{ C}$	A1
7(b)(ii)	$9.0 \times 10^{-8} \text{ C}$	A1
7(b)(iii)	capacitance = $(9.0 \times 10^{-8}) / 120$	C1
	$= 7.5 \times 10^{-10} \text{ F}$	A1
7(c)	total capacitance is halved	B1
	current is halved	B1

21.

6(a)	capacitance = charge / potential	M1
	charge is (numerically equal to) charge on one plate	A1
	potential is potential difference between plates	A1
6(b)(i)	two in series, in parallel with the other (correct symbols)	A1
6(b)(ii)	two in parallel connected to one in series (correct symbols)	A1
6(c)(i)	capacitance = $1.2 \mu\text{F}$	A1
6(c)(ii)	1. $Q = CV$	C1
	$= 1.2 \times 8.0$	A1
	$= 9.6 \mu\text{C}$	
	2. $E = \frac{1}{2}QV$ and $V = Q/C$ or $E = \frac{1}{2}CV^2$ and $V = Q/C$ or $E = \frac{1}{2}Q^2/C$	C1
	$E = \frac{1}{2}(9.6 \times 10^{-6})^2 / (3.0 \times 10^{-6})$ $= 1.5 \times 10^{-5} \text{ J}$	A1

22.

6(a)	Any valid two points e.g.: - to store (electrical) energy - smoothing/reduce ripple (on direct voltages/currents) - to block d.c. - timing/time delay (circuits) - in oscillator (circuits) - in tuning (circuits) - to prevent arcing/sparks	B2
6(b)	clear indication of equal charge on each capacitor	B1
	$E = V_1 + V_2 + V_3$ and $V = Q/C$	M1
	completion of algebra leading to $1/C = 1/C_1 + 1/C_2 + 1/C_3$	A1
6(c)(i)	three capacitors connected in parallel	B1
6(c)(ii)	parallel combination of three capacitors connected in series with one capacitor	B1

23.

6(a)(i)	charge per unit potential (difference)	M1
	charge on one plate <u>and</u> potential difference across the plates	A1
6(a)(ii)	any three points from: - smoothing - timing/(time) delay - tuning - oscillator - blocking d.c. - surge protection - temporary power supply	B3
6(b)	(capacitors in series have combined capacitance =) $8 \mu\text{F}$	C1
	capacitance = $8 + 24$ $= 32 \mu\text{F}$	A1

24.

6(a)(i)	charge per unit potential (difference)	M1
	charge on one plate and potential difference between the plates	A1
6(a)(ii)	any three points from: - smoothing - timing/(time) delaying - tuning - oscillator - blocking d.c. - surge protection - temporary power supply	B3
6(b)(i)	parallel combination of two in series and a single capacitor	B1
6(b)(ii)	one capacitor in series with two in parallel	B1

25.

7(a)	charge / potential	M1
	charge is on one plate, potential is p.d. between the plates	A1
7(b)(i)	$I = Q / t$	M1
	charge = CV <u>and</u> time = $1 / f$ leading to $I = fCV$	A1
7(b)(ii)	$4.8 \times 10^{-6} = 150 \times 60 \times C$	C1
	$C = 530 \text{ pF}$	A1
7(c)	(total) capacitance is halved	B1
	charge (for each cycle/discharge) is halved or since f and V are constant, current is proportional to capacitance	B1
	current = $2.4 \mu\text{A}$	B1

26.

6(a)	potential difference applied between the plates	M1
	causes charge separation (between the plates) or causes energy to be stored (between the plates)	A1
6(b)(i)	$I = Q / t$	M1
	clear substitution of $Q = CV$ and $f = 1 / t$, leading to $I = fCV$	A1
6(b)(ii)	$2.5 \times 10^{-6} = 50 \times C \times 180$	C1
	$C = 280 \text{ pF}$	A1
6(c)	(total) capacitance increases	B1
	greater charge (for each cycle/discharge) so greater (average) current or V and f are constant so (average) current increases or I is (directly) proportional to C so (average) current increases	B1

27.

6(a)	$Q = CV$ and $E = \frac{1}{2}CV^2$	B1
6(b)(i)	$C_N = CL / (L - D)$	B1
6(b)(ii)	(charge is unchanged by moving the plates so) $Q_N = CV$	B1
6(b)(iii)	$V_N = Q_N / C_N$ $= (CV) / [CL / (L - D)]$ $= V(L - D) / L$	B1
6(c)	oppositely charged plates attract, so energy stored decreases	B1

28.

5(a)	(energy stored =) area under line or $\frac{1}{2} QV$	C1
	$= \frac{1}{2} \times 8.0 \times 1.2 \times 10^{-4}$	
	$= 4.8 \times 10^{-4} \text{ J}$	A1
5(b)(i)	$(\tau =) RC$	C1
	$(\tau =) 220 \times 10^3 \times (1.2 \times 10^{-4} / 8.0) = 3.3 \text{ s}$	A1
5(b)(ii)	$E \propto V^2$	C1
	(so time to) $V_0 / 3$	C1
	$V = V_0 e^{-t/RC}$	
	$\frac{V_0}{3} = V_0 e^{-t/3.3}$	C1
	$\frac{1}{3} = e^{-t/3.3}$	
5(c)	(total) capacitance is doubled	M1
	time constant is doubled	A1

29.

5(a)	charge / potential (difference)	M1
	charge is charge on one plate, <u>and</u> potential is p.d. across the plates	A1
5(b)	p.d. across both capacitors = E	B1
	$Q_T = Q_1 + Q_2$	B1
	$C_T E = C_1 E + C_2 E$ hence $C_T = C_1 + C_2$	B1
5(c)(i)	$[(1/22) + (1/47)]^{-1} = 15 \mu\text{F}$	A1
5(c)(ii)	energy = $\frac{1}{2}CV^2$	C1
	$= \frac{1}{2} \times 15 \times 10^{-6} \times 12^2$	A1
	$= 1.1 \times 10^{-3} \text{ J}$	
5(c)(iii)	initial p.d. (across $22 \mu\text{F}$) = $12 \times (15/22)$	C1
	$= 8.2 \text{ V}$	
	or	
	final p.d. across both capacitors = $6.0 \times (22/15)$	
	$= 8.8 \text{ V}$	

$V = V_0 \exp [-t / (2.7 \times 10^6 \times 15 \times 10^{-6})]$	C1
$6.0 = 8.2 \exp [-t / (2.7 \times 10^6 \times 15 \times 10^{-6})]$ or $8.8 = 12 \exp [-t / (2.7 \times 10^6 \times 15 \times 10^{-6})]$ $t = 13 \text{ s}$	A1

30.

5(a)(i)	$Q = CV$	C1
	$Q_0 = 24 \times 470 \times 10^{-6}$ $= 0.011 \text{ C}$	A1
5(a)(ii)	$I_0 = 24 / 5600$ $= 4.3 \times 10^{-3} \text{ A}$	A1
5(a)(iii)	$\tau = RC$	C1
	$= 5600 \times 470 \times 10^{-6}$ $= 2.6 \text{ s}$	A1
5(a)(iv)	line with negative gradient throughout passing through (0, I_0)	B1
	exponential decay curve asymptotic to t -axis	B1
5(b)(i)	current in wire P gives rise to a magnetic field	B1
	as current (in P) changes, wire Q cuts (magnetic) flux (of wire P)	B1
	cutting magnetic flux causes induced e.m.f. (across Q)	B1
5(b)(ii)	sketch shows line with a negative gradient throughout	B1

31.

6(a)	- p.d. across resistor = p.d. across capacitor - current (in resistor) proportional to p.d. across it - current causes capacitor to lose charge - charge (on capacitor) proportional to p.d. so p.d. decreases <i>Any two points, 1 mark each</i>	B2
	rate of change of p.d. decreases as p.d. decreases	B1
6(b)	$Q_0 = 0.90 \text{ mC}$ and at $t = \text{one time constant}$, $Q = Q_0 \exp (-1)$	B1
	at $t = \text{one time constant}$, $Q = 0.90 \exp (-1) = 0.33 \text{ mC}$	M1
	evidence of graph reading: when $Q = 0.33 \text{ mC}$, $t = 5.5 \text{ s}$	A1
	or	
	evidence of two correct sets of readings for Q and t from the graph	(B1)
	correct substitution of Q and t values into $Q_2 = Q_1 \exp [(t_1 - t_2) / \tau]$	(M1)
	calculation to give $\tau = 5.5 \text{ s}$	(A1)
	or	
	read-off of half-life as 3.75 s	(B1)
	use of $Q = Q_0 \exp (-t / \tau)$ to show that $\tau = \text{half-life} / \ln 2$	(M1)
	$\tau = 3.75 / \ln 2 = 5.4 \text{ s}$	(A1)

6(b)	or	
	tangent drawn on $Q-t$ graph and value of Q at exact same time as tangent read from graph	(M1)
	gradient of tangent correctly calculated	(A1)
	$\tau = Q / \text{gradient}$ used to correctly calculate a value for τ as 5.5 s	(A1)
6(c)(i)	$C = Q / V$	C1
	$= [(0.90 \times 10^{-3}) / 7.5] = 1.2 \times 10^{-4} \text{ C}$	A1
	$= 120 \mu\text{F}$	
6(c)(ii)	$R = \tau / C$	C1
	$= 5.5 / (1.2 \times 10^{-4}) (= 45\,800 \Omega)$	A1
	$= 46 \text{ k}\Omega$	

32.

5(a)	from graph $\ln Q = 2.9$	B1
	(so $Q = 18.2 \mu\text{C}$)	
	$C = Q / V$	C1
	$= 18.2 / 12 = 1.5 \mu\text{F}$	A1
5(b)	gradient = -0.25	C1
	gradient = $-1 / RC$	C1
	$R = 1 / (0.25 \times 1.5 \times 10^{-6})$	A1
	$= 2.7 \times 10^6 \Omega$	
	or $\frac{Q}{Q_0} = e^{-t/CR}$ or $\ln Q - \ln Q_0 = \frac{-t}{CR}$	(C1)
	e.g. $\frac{4.95}{18.2} = e^{-5.2 / (1.5 \times 10^{-6} R)}$ or $1.6 - 2.9 = 5.2 / (1.5 \times 10^{-6} R)$	(C1)
5(c)	$R = 2.7 \times 10^6 \Omega$	(A1)
	$W = \frac{1}{2} QV$	C1
	$= \frac{1}{2} \times 18.2 \times 10^{-6} \times 12$	A1
	$= 1.1 \times 10^{-4} \text{ J}$	
	or $W = \frac{1}{2} CV^2$	(C1)
	$= \frac{1}{2} \times 1.5 \times 10^{-6} \times 12^2$	(A1)
	$= 1.1 \times 10^{-4} \text{ J}$	
5(d)	or $W = \frac{1}{2} Q^2 / C$	(C1)
	$= \frac{1}{2} \times (18.2 \times 10^{-6})^2 / 1.5 \times 10^{-6}$	(A1)
	$= 1.1 \times 10^{-4} \text{ J}$	
	straight line with different negative gradient starting from (0, 2.9)	M1
	straight line between $t = 0$ and at least $t = 5.0\text{s}$ with twice the gradient of the original line	A1

33.

5(a)(i)	$Q_A = CV$	A1
5(a)(ii)	$E_A = \frac{1}{2}CV^2$	A1
5(b)(i)	some of the charge transfers to (the plates of) capacitor B	B1
	transfer is because the p.d.s across the capacitors are not equal or transfer stops when the p.d.s across the capacitors become equal	B1
5(b)(ii)	$V_A = V_B$	M1
	charge on A + charge on B = CV	M1
	$CV_B + 3CV_B = CV$ leading to $V_B = V/4$	A1
	or	
	$C_T = 4C$	(M1)
	$Q_T = CV$	(M1)
	$V_B = CV/4C = V/4$	(A1)
5(b)(iii)	$\Delta E = \frac{1}{2}CV^2 - nCV^2$, where n is a multiple that is less than $\frac{1}{2}$ or total final energy = $\frac{1}{2} \times 4C \times (V/4)^2$ = $\frac{1}{8}CV^2$	C1
	$\Delta E = \frac{1}{2}CV^2 - \frac{1}{8}CV^2$ = $\frac{3}{8}CV^2$	A1

34.

6(a)(i)	energy stored = area under graph	C1
	= $\frac{1}{2} \times 450 \times 10^{-6} \times 8.0 = 1.8 \times 10^{-3} \text{ J}$ or 1.8 mJ	A1
6(a)(ii)	$C = Q/V$ or $E = \frac{1}{2}CV^2$	C1
	$C = (450 \times 10^{-6})/8.0$ or $(2 \times 1.8 \times 10^{-3})/8.0^2$ = $5.6 \times 10^{-5} \text{ F}$	A1
6(b)(i)	$V = V_0 \exp(-t/RC)$ and $\tau = RC$	C1
	$V = V_0 \exp(-t/\tau)$	A1
	$V_0 = 8.0 \text{ V}$, and at one time constant, $t = \tau$	
	$V/8.0 = \exp(-\tau/\tau)$, so $\ln(V/8.0) = -1.0$ or $-\ln(V/8.0) = 1.0$	
6(b)(ii)	[t read from graph at $-\ln(V/8.0) = 1.0$]: $\tau = 3.2 \text{ s}$	A1
6(b)(iii)	$\tau = RC$	C1
	$R = 3.2 / (5.6 \times 10^{-5})$ = $5.7 \times 10^4 \Omega$	A1